

Pedestrian Occlusion Detection and Tracking Method Based on Improved PP-YOLOE+DeepSORT algorithm

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Abstract—This paper proposes a pedestrian target detection and tracking algorithm based on the improved PP-YOLOE and DeepSORT. It aims to address the problems of low accuracy and poor robustness of traditional target detection and tracking algorithms in scenes where human bodies occlude each other. In the detection part, the spatial attention mechanism is applied to the PP-YOLOE algorithm so that the model can focus on key regions in the image and extract more expressive features. In addition, this paper's network model is improved to reduce the computational complexity and speed up the detection speed. In the tracking part, we choose the DeepSORT algorithm to track the detected pedestrian features in real time. Finally, experimental results show that the model's design and improvement can accurately detect and track pedestrians in complex environments with occlusion and high target density.

Keywords—Object detection, Target tracking, PP-YOLOE, Attention mechanism, Depthwise Separable Convolution

I. INTRODUCTION

In recent years, pedestrian detection and tracking have become a hot research direction in computer vision and artificial intelligence and have many practical uses, such as video surveillance and automotive-assisted driving [1, 2]. However, in practical application scenarios, the complexity of the environment, such as dynamic background and occlusion, and the constant change of human body posture during pedestrian movement increase the difficulty of pedestrian detection and have high requirements on the speed and robustness of the algorithm's processing.

Traditional target detection methods usually focus on analysing local features of an image, such as shape and texture. However, their performance and adaptability could be better when dealing with global contextual information. Deep learning-based target detection methods effectively compensate for the shortcomings of traditional methods, which can be classified into single-stage and two-stage detection methods. These methods can handle complex image features and improve the accuracy and efficiency of target detection. The R-CNN series algorithms are one of the two-stage methods that extract candidate regions before precise localization and classification with high accuracy but high computational cost [3-6]. Single-stage methods, such as the YOLO series and SSD [7-10], directly predict the whole image by simplifying the process, whose speed is fast and the process is concise. With the development of the technology, anchor-free and single-stage target detection algorithms have emerged. These deep learning algorithms abandon the reliance on predefined anchor frames in target detection and

directly predict the location in the image where the detected target is located and its class. For example, CenterNet focuses on the centre of the location of the projected target [11]; FCOS directly predicts the bounding box and class of each pixel point using the complete convolution method [12]; and the PP-YOLOE combines the idea of the FCOS by placing an anchor point at each pixel point to place anchor points, which improves the flexibility and accuracy of detection [13]. These anchor-free methods have significantly progressed in localization accuracy, inference speed, and handling of small targets and dense scenes. In addition, the algorithm is more deployment-friendly as it is structured to avoid using operators that are not easy to deploy, such as deformable convolution and matrix nms.

This paper comprehensively adopts the PP-YOLOE single-stage target detection network based on an anchorless frame for human target detection. Meanwhile, considering the multi-target characteristic of detection in complex backgrounds, this paper will apply a multi-target tracking algorithm for human body tracking.

II. DESIGN OF OVERALL ALGORITHM STRUCTURE

A. Description of PP-YOLOE Object Detection Method

The PP-YOLOE algorithm has a much higher detection speed and accuracy balance than other previously released algorithms in the YOLO series. The PP-YOLOE algorithm has four models (s/m/l/x) to adapt to different application needs. Considering the network model's scale and detection time cost, this paper chooses PP-YOLOE-S to construct the target detection model. Fig. 1 shows the structure of the PP-YOLOE algorithm, which contains three parts: the backbone, neck, and head.

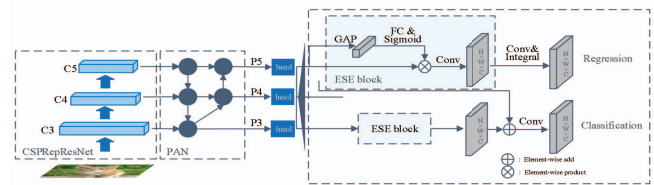


Fig. 1. PP-YOLOE structure.

First, the backbone layer is crucial to extracting the feature information in target detection. The backbone layer in the PP-YOLOE is the CSPRepResNet structure. Secondly, the neck layer is the intermediate layer of the object detection network. It is responsible for fusing features to the feature map and transmitting them to the prediction layer. The Neck layer of PP-YOLOE is the Custom CSPPAN network, which

utilizes the Feature Pyramid Network and Path Aggregation Network architecture. Finally, the primary function of the head layer is to process the feature pyramid to detect the target. The PP-YOLOE algorithm's head layer employs a head based on the TOOD network and replaces the layer attention in the T-head with a lightweight ESE (Effect Squeeze and Extract) to form the ET-head [14].

B. Improvement of PP-YOLOE

We integrate the spatial attention mechanism into the backbone layer, strengthen the model's sensitivity to specific spatial locations, improve target detection accuracy, and reduce complex background interference. Meanwhile, we improve the network structure of PP-YOLOE by introducing the decoupled head and depthwise separable convolution. These approaches simplify the computations and speed up the detection speed.

1) *Spatial Attention Mechanism*: The attention mechanisms mimic the human attention process by selectively focusing on some parts of the information and ignoring others, thus enabling the model to extract more meaningful information. The attention mechanisms can be categorised into channel attention and spatial attention. The channel attention focuses on the importance of different channels in the channel dimension of the feature graph. The spatial attention, on the other hand, focuses on the spatial dimension of the feature graph and learns to allocate attention at each spatial location of the feature graph [15]. The ESE module of PP-YOLOE includes channel attention mechanisms. Therefore, we introduce the spatial attention mechanism to enable PP-YOLOE to accurately identify important information in images under occlusion and complex backgrounds. Fig. 2 shows the Channel-Spatial attention structure.

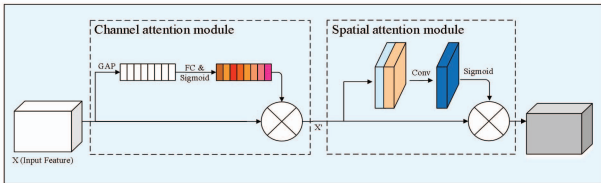


Fig. 2 Channel-Spatial attention structure.

First, in the channel attention module, a global average pooling operation computes the average value of each channel in the feature map. Secondly, the channel attention module learns the weights of each channel through one or more Full Connections and generates the channel weights through the sigmoid activation function. Finally, the learned channel weights are multiplied by the initial input feature map to recalibrate the features. This step reinforces those channels considered necessary by the model and suppresses unimportant channels. The calculation process is shown in Equation (1-3).

$$g(X) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j) \quad (1)$$

$$s = \sigma(FC(g(X))) = \sigma(Conv(g(X))) \quad (2)$$

$$X' = F_{scale}(X, s) = X \otimes s \quad (3)$$

where X denotes the input feature map, H and W denote the height and width of the feature map, σ denotes the Sigmoid activation function, $g(X)$ denotes the global average pooled

vector, $Conv$ denotes the convolution operation, and $\otimes$ denotes the element-by-element product.

In the spatial attention module, the input X' is first passed through a convolutional layer to capture local spatial dependencies; secondly, the activation function Sigmoid normalises the output of the convolutional layer into a spatial attention map; finally, the feature map X' is multiplied element by element with S_1 to obtain the weighted feature map $\hat{X}$. Equation (4-5) illustrates the computation method.

$$S_1 = \sigma(Conv(X')) \quad (4)$$

$$\hat{X} = X' \otimes S_1 \quad (5)$$

Adding the spatial attention mechanism enables the model to perform more detailed feature processing and focus on key regions in the image. In scenes where occlusion exists, the spatial attention mechanism can help the model focus on unoccluded regions, thereby improving target detection accuracy. In complex backgrounds, spatial attention can also help the model distinguish between foreground targets and background noise.

2) *Module lightweighting* a). *Decoupled Head*: The head of conventional target detection models typically manages the regression and classification tasks. The decoupling header completely separates the classification task from the regression task to improve efficiency and accuracy. The classification task focuses on the features that distinguish the target from the background. In contrast, the regression task focuses on the target's spatial attributes, such as the position and size of the bounding box [16].

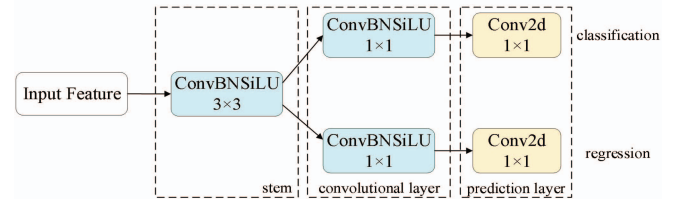


Fig.3. Decoupled head structure.

The head of PP-YOLOE target detection replaces the layer attention in the T-head with lightweight ESE attention to form the ET-head. However, ET-head reduces the inference speed in the model and has almost no accuracy gain. Therefore, as shown in Fig. 3, we replace the ET-head with an efficient decoupling header. The goal of the stem layer is to perform preliminary processing of the input feature maps in preparation for subsequent classification and regression tasks. It consists of a convolutional layer, sigmoid linear unit (SiLU), and batch normalization (BN). Meanwhile, the model extracts features for classification and regression through the convolutional layer. Finally, the model separately predicts each anchor's category scores and bounding box coordinates through the prediction layer.

b) Depthwise Separable Convolution

The depthwise separable convolution lowers the computational cost and parameter count in convolutional neural networks. Suppose the size of the input feature map is $C_{in} \times H \times W$, and the convolution is performed in three groups. When performing an ordinary convolution operation, the quantity of channels in each set of output feature maps is $C_{out}/3$, and the size of each convolution kernel is $C_{in} \times K \times K$. Connecting the results of the convolution calculation of

each group outputs C_{out} feature maps. Therefore, the parameter count in the standard convolution is $C_{in} \times K \times K \times C_{out}$.

After implementing the depthwise separable convolutional structure, we analysed the computing complexity as follows: firstly, each convolution kernel in the depthwise convolution is responsible for one channel; that is, each channel passes through a convolution kernel of size $K \times K$ to get the output feature maps. Thus, the depthwise convolution section has $C_{in} \times K \times K$ parameters. Secondly, the pointwise convolution resembles ordinary convolution, with a kernel size of $C_{in} \times 1 \times 1$. The pointwise convolution generates a new feature map by interacting between channels with the number of parameters $C_{in} \times 1 \times 1 \times C_{out}$.

The analysis presented above indicates that there are $C_{in} \times K \times K \times C_{out}$ parameters in a regular convolution and $C_{in} \times K \times K \times C_{out} + C_{in} \times 1 \times 1 \times C_{out}$ parameters in the depthwise separable convolution. The model's parameters have been significantly decreased.

C. Multi-object Tracking

The DeepSORT algorithm is an advanced technique for real-time multi-target tracking that consists of three main components: state estimation, matching problems, and trajectory management [17].

Firstly, state estimation is a critical step in multi-target tracking for predicting the motion state of each target in a video sequence. The DeepSORT algorithm adopts the Kalman filter as its state estimator, which forecasts the condition of the target in the next frame using the dynamic model of the target and updates the state in conjunction with the observation data. Secondly, the DeepSORT algorithm uses the Hungarian algorithm to resolve the matching problem and introduces a multidimensional feature space to enhance the matching accuracy. Finally, in the trajectory management section, the DeepSORT algorithm will create a new trajectory for detection if it fails to match an existing trajectory, update the status of an existing trajectory based on the detection if the detection is successfully associated with an existing trajectory, and delete a trajectory if it has not been associated with any detection for a certain period. By working with these three components, the DeepSORT algorithm can provide accurate tracking performance in complex multi-target tracking scenarios. In addition, by introducing deep features, the DeepSORT algorithm maintains high tracking accuracy even when the target appearance changes significantly.

III. EXPERIMENTS

The experimental environment used for human target detection and tracking is the Ubuntu operating system, PaddlePaddle 2.4.0 deep learning framework, Python 3.8, AMD EPYC 7352 24-Core Processor, and NVIDIA GeForce RTX 3090 GPU.

A. DATASET INTRODUCTION

The target detection part of the experiments in this paper is trained and validated on the Crowdhuman pedestrian dataset. This publicly available dataset contains scenes of severe pedestrian occlusion caused by high congestion levels. It contains many high-quality images and annotated information. We use 15000 images as the training dataset for the experiment, 4350 images as the validation dataset, and 5018 images as the test dataset. In addition, we use the MOT16 dataset for the target tracking part of the test. The MOT16 is a standard target-tracking dataset proposed in 2016 to measure multi-target tracking methods.

B. Results and Data Analysis

Since the default parameters of PP-YOLOE are for multi-category target detection, and this paper only needs to detect the human body, we have adjusted some of the parameters. The detection category is adjusted to 1, the class_loss ratio is increased appropriately, and some parameters are adjusted. Table 1 shows the comparison results before and after the parameter adjustment.

TABLE I. RESULT AFTER PARAMETER ADJUSTMENT

model	mAP(%)	MOTA
Before tuning the parameters	80.31	37
After tuning the parameters	79.76	65

We perform ablation tests for every improvement point to confirm the performance of the enhanced PP-YOLOE algorithm. Table 2 displays the experiment results. After adding the spatial attention mechanism to the network, the target detection accuracy increases to 82.06%, but the speed decreases slightly. After introducing the decoupling head in the original network, the detection accuracy remains stable, and the detection speed is significantly improved. However, target detection accuracy is significantly reduced after lightweighting all convolutional modules on this basis. So, we adapted the depth-separable convolution proportion and conducted a series of tests. The balance of accuracy and speed is achieved when about 1/5 of the regular convolution is replaced with the depth-separated convolution. After introducing the spatial attention mechanism and the lightweight module, the network achieved 81.10% accuracy and 72 fps speed.

TABLE II. RESULT OF THE ABLATION EXPERIMENT

model	spatial attention mechanism	module lightweighting	mAP(%)	FPS
PP-YOLOE	×	×	79.76	65
PP-YOLOE+A	√	×	82.06	62
PP-YOLOE+B	×	√	78.20	76
PP-YOLOE+A+B	√	√	81.10	72



Fig. 4. Comparative results.
(a) Detection results before network optimisation, (b) Detection results after network optimisation.

We compared the original network with the augmented network to visualize the algorithm's improvement, as shown in Fig. 4. The improved network performs better in occlusion situations, detects occluded body parts more accurately, and has a higher confidence level in detecting occluded bodies.

The human body information detected by the improved PP-YOLOE is input into the target tracking algorithm to complete the tracking of pedestrians. This paper conducts training tests of several algorithms on the MOT16. Table 3 displays that the mAP scores of the three multi-target tracking algorithms are not significantly different. However, DeepSORT has the highest MOTA score, which indicates that it has higher overall accuracy in tracking the object, can better maintain the target's identity, and reduces tracking interruptions and mistracking.

TABLE III. COMPARISON RESULTS OF MULTI-TARGET TRACKING ALGORITHMS ON MOT16

algorithm	mAP(%)	MOTA
ByteTrack [18]	52.9	51.1
BOT-SORT [19]	52.7	55.9
DeepSORT	52.9	57.5

As shown in Fig. 5, we combined the improved PP-YOLOE with DeepSORT to perform target tracking on two videos in public places to verify the algorithm's practical application. The resultant graph shows that even though the pedestrians are occluded, they are still accurately recognized and continuously tracked, and their ID numbers remain consistent.



Fig. 5. PP-YOLOE-DeepSORT pedestrian tracking experiment.

IV. CONCLUSION

This paper discusses target detection and tracking in pedestrian occlusion scenarios, including the basic structure

of the PP-YOLOE algorithm, the addition of spatial attention mechanisms and lightweight modules in the model to improve PP-YOLOE, and the selection of multi-target tracking algorithms. Experiments show that the pedestrian detection and tracking algorithm based on the improved PP-YOLOE and DeepSORT algorithms achieves good accuracy and real-time results. Meanwhile, combining these two algorithms for pedestrian detection and tracking can effectively exclude the influence of the complex environment, such as occlusion, and enhance detection accuracy and tracking stability. However, the system has unavoidable drawbacks, such as detection delay when deployed to real complex scenes. In our subsequent work, we will further improve the pedestrian detection system for better performance and higher detection accuracy in heavily occluded and complex scenes.

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